



David Robert Grimes PhD, CStat is a scientist, broadcaster, and author. An expert in the public understanding of medical science, he has contributed to outlets including the *Guardian*, *New York Times*, BBC, and *Financial Times*. He was joint recipient of the 2014 John Maddox Prize and is author of the book *The Irrational Ape* (Simon & Schuster, 2019).

Bad stats

A regular series exploring slip-ups, snafus and salutary lessons from the world of statistics

Promiscuous dichotomisation

No, 5G did not cause Covid, and sugar does not give you herpes.
David Robert Grimes slams data torture and the madness of arbitrary thresholds

The economist Ronald Coase famously warned that if one tortures the data, it will confess to anything. In biomedical science, there are alarming signs that this lesson has not been understood. While we still do not know the full extent of the problem, what we do know is concerning: only 11% of landmark preclinical trials replicate in oncology.¹ A seminal Alzheimer's research paper took 18 years to retract, siphoning funding and time from more promising ideas and ultimately harming patients (tinyurl.com/y7n2cn7a). Covid-related publishing was defined by high volumes of poor methodological quality (tinyurl.com/ykkmd6mh) while approximately 85% of medical literature is deemed poor-quality or spurious, of minimal use to clinicians, policy-makers, and patients, a problem dubbed "the scandal of research waste".²

There are numerous reasons for this sorry state of affairs, and understanding the reasons why poor research emerges is important to counteract the worst excesses of untrustworthy research before it harms patients. To this end, we recently looked at a scenario that is all too easy to engineer. We considered a form of what is effectively statistical gerrymandering, where a cunning operator can simply redraw arbitrary boundaries in data to conjure up whatever trend they so desire. It is inherently a form of undeclared multiple testing, and the magic of the threshold is that a less than diligent experimenter can justify its selection *post hoc*, masking the fact that they have extracted a data confession under conditions of torture.

To understand how this manifests, it is informative to consider some examples.

In the heady early days of the pandemic, claims that 5G frequency radiation caused Covid-19 became inexplicably popular. As I had covered the rise of the anti-5G movement prior to the pandemic in my science communication capacity, I was invited to appear on a BBC programme debunking this claim in April 2020. That appearance and my comments drew the ire of a lobby group convinced of hidden harms from Wi-Fi and cellular phones. They sent a long diatribe to a BBC journalist, consisting of a screed of seemingly legitimate academic publications they insisted supported precisely their position. Anxious to check this, the BBC journalist sent the packet to me for appraisal. Most were either deeply misconstrued or low-quality anecdotal claims, but one curious paper from late 2017 deserved some attention. In this publication, the authors claimed to have traced the magnetic field exposure of 913 women, finding that high exposure was associated with increased risk of miscarriage. In the associated press release, the authors had made clear their contention that cellular phones and Wi-Fi routers were a clear risk to pregnant women.

Having spent my early research career in both ionising and non-ionising radiation, this was a shocking claim which I was somewhat sceptical of for a multitude of reasons. To arrive at their conclusion, the authors defined a threshold for "high" exposure, and then counted miscarriages above and below this threshold. This was not the first time they had

taken such an approach, nor the last – in other papers, they had done precisely the same thing for other conditions, again setting a threshold for high exposure and then counting the incidence of conditions like ADHD and asthma above and below this threshold. What was curious was that these thresholds had no clear physical significance, nor consistency; in the miscarriage paper, they collapsed the entirety of a woman's daily exposure level to a single metric, the 99th percentile of her daily exposure. They then deemed "high" exposure as levels above the first quartile of the resultant distribution. This seemed a rather strange designation, especially as in other papers they had used a host of different definitions for high exposure.

Working backwards from the summary statistics in the paper revealed some problems, including basic data anomalies, and the fact they had jettisoned over half the recorded subjects for apparent non-representativeness. But most concerning was the immediate and obvious problem that their threshold seemed rather arbitrary. Concerned about the veracity of their approach, I contacted my friend and collaborator James Heathers with my concerns. James has something of a reputation for detecting problems in scientific literature, a self-described "data thug" with a gift for revealing fatal issues in bold scientific claims (tinyurl.com/f422wa78). Between us, we enumerated a slew of problems with not just the miscarriage paper, but all of the similar papers by the group, and found evidence that the arbitrary nature of the thresholds selected drove the apparent results. How journals responded varied as it always does; one quietly retracted the papers in question, while others completely ignored the issues raised. One editor invited us to write a short research report on what we had found.³ When the authors refused to supply their raw data for scrutiny, the miscarriage publication earned

We considered a form of what is effectively statistical gerrymandering, where a cunning operator can simply redraw arbitrary boundaries in data to conjure up whatever trend they so desire

That researchers should not dichotomise continuous variables is a piece of statistical advice often given and just as frequently ignored

itself an editorial expression of concern in 2021, four years after its initial debut.

That was but one example of the problem of arbitrary thresholds, ubiquitous across biomedical science. Vitamin D research is an exemplar of precisely this. In the early 2000s, the humble hormone seemed on the brink of a revolution. Early research hinted that low levels could drive everything from cancer to cardiovascular disease, dementia to depression. An explosion of popular books extolled the virtues of vitamin D as a miraculous panacea. And throughout the world, the message that we were not getting enough constantly percolated aggressively. In one *Good Morning America* segment, the reporter Diane Sawyer insisted that 100 million Americans were deficient. Vitamin D testing and supplementation became a roaring market worldwide. A quarter of a century later, the picture is markedly different. In stark contrast to the initial excitement, a host of better-quality experimental data has shown absolutely no impact of vitamin D on cancer or heart disease risk. Nor did amplifying intake prevent falls, improve cognitive function or reduce the risk of bone fracture. And despite the solemn intonation that vast swathes of the world were deficient, controlled population testing showed that this was based on a faulty assumption over what constituted a deficiency.

Part of the problem was that even now, the threshold for deficiency is bizarrely *ad hoc*. There are at least seven different thresholds used in literature, ranging from 8 ng/ml all the way up to 30 ng/ml. An investigation in *Scientific American* ([tinyurl.com/3w8zcdnh](https://www.sciencemag.org/3w8zcdnh)) found that some of the confusion arose from conflicting standards. In 2011, an expert panel from the Institute of Medicine (IOM) reviewed all then existent vitamin D research and concluded that the apparent benefits plateaued around 12–16 ng/ml, with no benefits to having levels above 20 ng/ml. The same year, the Endocrine Society pushed a guideline deeming levels below 20 ng/ml as “deficient” and created a new range (20–29 ng/ml) they dubbed “insufficient”. Immediately this proved controversial, criticised roundly

by the IOM. Confusing matters further was the fact that the then chair of the Endocrine Society had a clear conflict of interest, having authored a book extolling the virtues of high-dose vitamin D and with financial ties to companies selling testing and supplements. While the Endocrine Society quietly revised this downwards in 2024, the existence of multiple deficiency thresholds means that researchers can effectively pick whichever level that most accentuates a desired result.

This is a problem we dubbed “promiscuous dichotomisation”, and ascertaining its impact became a motivating focus. That researchers should not dichotomise continuous variables is a piece of statistical advice often given and just as frequently ignored. The reason typically cited for this being bad practice is that it greatly reduces one’s statistical power, or a researcher’s ability to detect an effect when one exists.^{4,5} But we were motivated by a slightly different question – instead of false negatives, how likely are we to erroneously arrive at a false positive if we allow a researcher to make any arbitrary cut in the data they desire? While this is intimately related to multiple testing, it is not a trivial question to answer because the data remains unchanging, so these are not simply independent tests. But it is amendable to some exact solutions, and to simulation when analytical approaches become unwieldy due to the sheer volume of combinatorics involved.

What we found was concerning: in data where no real pattern existed, the probability of obtaining a false positive result depended on the number of individuals involved and the proportion with a given outcome. The greater the number of individuals, the greater the probability that one could obtain a false positive result by simply picking a suitable threshold to turn noise into apparent signal. Even with just 100 participants, it was possible to get over a 38% probability of a false positive result, rising to well over 66% chance if low count scenarios were not excluded. By the time one has 1,000 participants, this rises to well over 80% probability of a false positive result in any arbitrary set of data. To

illustrate this, we took the National Health and Nutrition Examination Survey (NHANES) and considered a continuous measure (blood glucose) and a binary outcome utterly unrelated to it, herpes status. Considering all data, a logistic regression finds no relationship between the twain, as expected – but given the freedom to dichotomise the continuous variable, it was trivial to find an arbitrary threshold that gave the desired illusion. In our case, simply designating 4.3 mmol/l as the threshold for “high” glucose gave a spurious apparent hazard ratio of 2.04 for herpes type II in the “high” group relative to the “low” one.

This entirely ridiculous illustration gave us the title for our work, “‘Cake causes herpes?’ – Promiscuous dichotomisation induces false positives”.⁶ But deliberately outlandish as our example was, it is unfortunately informative. Due to the ease with which researchers can now torture data, there has been an explosion of subpar and formulative research inappropriately mining databases like NHANES to arrive at speculative and spurious conclusions.⁷ Promiscuous dichotomisation is undoubtedly bad research practice, and results derived from it are inherently unreliable. We would do well to remember physicist Richard Feynman’s advice to all scientists: “The first principle is that you must not fool yourself – and you are the easiest person to fool.” ■

References

1. Begley, C. and Ellis, L. (2012) Raise standards for preclinical cancer research. *Nature*, **483**, 531–533.
2. Glasziou, P. and Chalmers, I. (2018) Research waste is still a scandal. *British Medical Journal*, **363**, k4645.
3. Grimes, D. R. and Heathers, J. (2021) Association between magnetic field exposure and miscarriage risk is not supported by the data. *Scientific Reports*, **11**, 22143.
4. Riley, D. R., Cole, T. J., Deeks, J., Kirkham, J. J., Morris, J., Perera, P., Wade, A. and Collins, G. S. (2022) On the 12th day of Christmas, a statistician sent to me. *British Medical Journal*, **379**, e072883.
5. Royston, P., Altman, D. G. and Sauerbrei, W. (2006), Dichotomizing continuous predictors in multiple regression: A bad idea. *Statistics in Medicine*, **25**, 127–141.
6. Grimes, D. R. and Heathers, J. (2025) “‘Cake causes herpes?’ – Promiscuous dichotomisation induces false positives. *BMC Medical Research Methodology*, **25**, 255.
7. Suchak, T., Aliu, A. E., Harrison, C., Zwiggelaar, R., Geifman, N. and Spick, M. (2025) Explosion of formulaic research articles, including inappropriate study designs and false discoveries, based on the NHANES US national health database. *PLoS Biology*, **23**(5), e3003152.